

Week 7

Student Performance – Data Cleaning & Transformation

SANDHYA SHINDE S

1. Introduction

The Week 7 project focuses on Student Performance – Data Cleaning and Transformation using Microsoft Power BI and Power Query. The main purpose of this project is to clean, organize, and transform the student performance dataset so that it can be used for further analysis.

In this project, I imported the student performance dataset, inspected and renamed the columns, created Total Score and Average Score, classified students using Grade and Pass/Fail, performed Unpivot on the subject scores, and rounded the Average Score. This project helped me understand how raw data can be transformed into a structured and analysis-ready dataset using Power Query.

2. Import the Dataset

I imported the Student Performance in Exams dataset into Power BI using the Text/CSV option. After importing the dataset, I selected Transform Data to open the Power Query Editor.

StudentsPerformance.csv

File Origin: 1252: Western European (Windows) Delimiter: Comma Data Type Detection: Based on first 200 rows

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75
female	group B	associate's degree	standard	none	71	83	78
female	group B	some college	standard	completed	88	95	92
male	group B	some college	free/reduced	none	40	43	39
male	group D	high school	free/reduced	completed	64	64	67
female	group B	high school	free/reduced	none	38	60	50
male	group C	associate's degree	standard	none	58	54	52
male	group D	associate's degree	standard	none	40	52	43
female	group B	high school	standard	none	65	81	73
male	group A	some college	standard	completed	78	72	70
female	group A	master's degree	standard	none	50	53	58
female	group C	some high school	standard	none	69	75	78
male	group C	high school	standard	none	88	89	86
female	group B	some high school	free/reduced	none	18	32	28
male	group C	master's degree	free/reduced	completed	46	42	46
female	group C	associate's degree	free/reduced	none	54	58	61

The data in the preview has been truncated due to size limits.

Extract Table Using Examples Load Transform Data Cancel

3. Explore and Inspect the Data

I checked the columns and records available in the dataset. The dataset contains student information such as:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score

I also added an Index Column and named it StudentID.

Table: ReorderColumns(4"Renamed Columns",{"StudentID", "gender", "race/ethnicity", "parental level of education", "lunch", "test preparation course"}

StudentID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	female	group B	bachelor's degree	standard	none
2	female	group C	some college	standard	completed
3	female	group B	master's degree	standard	none
4	male	group A	associate's degree	free/reduced	none
5	male	group C	some college	standard	none
6	female	group B	associate's degree	standard	none
7	female	group B	some college	standard	completed
8	male	group B	some college	free/reduced	none
9	male	group D	high school	free/reduced	completed
10	female	group B	high school	free/reduced	none
11	male	group C	associate's degree	standard	none
12	male	group D	associate's degree	standard	none
13	female	group B	high school	standard	none
14	male	group A	some college	standard	completed
15	female	group A	master's degree	standard	none
16	female	group C	some high school	standard	none
17	male	group C	high school	standard	none
18	female	group B	some high school	free/reduced	none
19	male	group C	master's degree	free/reduced	completed
20	female	group C	associate's degree	free/reduced	none
21	male	group D	high school	standard	none
22	female	group B	some college	free/reduced	completed
23	male	group D	some college	standard	none
24	female	group C	some high school	standard	none
25	male	group D	bachelor's degree	free/reduced	completed
26	male	group A	master's degree	free/reduced	none
27	male	group B	some college	standard	none

9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 08:53

4. Rename the Columns

I renamed the columns to make them easier to understand and use.

For example:

- gender → Gender
- race/ethnicity → Race/Ethnicity
- parental level of education → Parental Level of Education
- lunch → Lunch
- test preparation course → Test Preparation Course
- math score → Math Score
- reading score → Reading Score
- writing score → Writing Score

The screenshot shows the Power BI Desktop interface. The main view displays a table named 'StudentsPerformance' with the following data:

StudentID	Gender	Race/Ethnicity	Parental level of Education	Lunch	Test Preparation Course
1	female	group B	bachelor's degree	standard	none
2	female	group C	some college	standard	completed
3	female	group B	master's degree	standard	none
4	male	group A	associate's degree	free/reduced	none
5	male	group C	some college	standard	none
6	female	group B	associate's degree	standard	none
7	female	group B	some college	standard	completed
8	male	group B	some college	free/reduced	none
9	male	group D	high school	free/reduced	completed
10	female	group B	high school	free/reduced	none
11	male	group C	associate's degree	standard	none
12	male	group D	associate's degree	standard	none
13	female	group B	high school	standard	none
14	male	group A	some college	standard	completed
15	female	group A	master's degree	standard	none
16	female	group C	some high school	standard	none
17	male	group C	high school	standard	none
18	female	group B	some high school	free/reduced	none
19	male	group C	master's degree	free/reduced	completed
20	female	group C	associate's degree	free/reduced	none
21	male	group D	high school	standard	none
22	female	group B	some college	free/reduced	completed
23	male	group D	some college	standard	none
24	female	group C	some high school	standard	none
25	male	group D	bachelor's degree	free/reduced	completed
26	male	group A	master's degree	free/reduced	none
27	male	group B	some college	standard	none

The right-hand pane shows the 'Query Settings' for 'StudentsPerformance'. The 'APPLIED STEPS' list includes: Source, Promoted Headers, Changed Type, Added Index, Renamed Columns, and Renamed Columns1 (which is currently selected).

5. Create Total Score

I created a Custom Column named Total.

The formula used was:

Total = Math Score + Reading Score + Writing Score

This calculates the total marks obtained by each student in all three subjects.

Custom Column

Add a column that is computed from the other columns.

New column name

Total

Custom column formula ⓘ

= [Math Score]+[Reading Score]+[Writing Score]

Available columns

Gender
Race/Ethnicity
Parental level of Education
Lunch
Test Preparation Course
Math Score
Reading Score
Writing Score

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

6. Create Average Score

I created another Custom Column named Average.

The formula used was:

Average = Total / 3

This calculates the average marks of each student across Math, Reading, and Writing.

Custom Column

Add a column that is computed from the other columns.

New column name

Average

Custom column formula ⓘ

= [Total]/3

Available columns

Parental level of Education
Lunch
Test Preparation Course
Math Score
Reading Score
Writing Score
Total

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

7. Create Grade

I created a Conditional Column named Grade based on the Average score.

The conditions used were:

- Average $\geq$ 90 → Grade A
- Average $\geq$ 75 → Grade B
- Average $\geq$ 50 → Grade C
- Average $\geq$ 40 → Grade D
- Below 40 → Fail

This helps to classify students according to their overall performance.

Table.TransformColumnTypes(#"Added Custom1",{{"Average ", type number}})

Test Preparation Course | Math Score | Reading Score | Writing Score | Total | Average

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name:

Condition	Column Name	Operator	Value	Output
If	Average	is greater than or...	90	Grade A
Else If	Average	is greater than or...	75	Grade B
Else If	Average	is greater than or...	50	Grade c
Else If	Average	is greater than or...	40	Grade D

Else:

Test Preparation Course	Math Score	Reading Score	Writing Score	Total	Average
completed	74	71	80	225	75
none	73	74	72	219	73
none	69	54	55	178	59.33333333

8. Create Pass/Fail

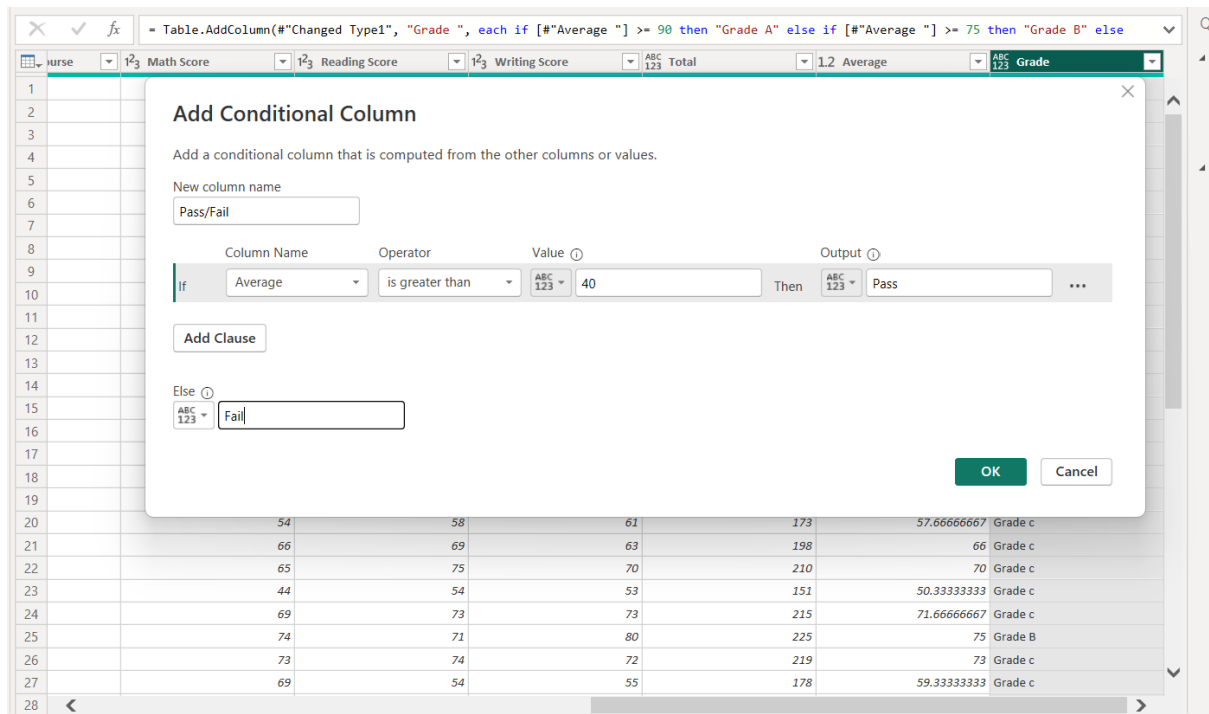
I created another Conditional Column named Pass/Fail.

The condition used was:

Average > 40 → Pass

Otherwise → Fail

This classifies students based on the defined passing criteria.



9. Unpivot Subject Scores

I selected the Math Score, Reading Score, and Writing Score columns and used the Unpivot Columns option in Power Query.

After unpivoting, the subject scores were transformed into two columns:

- Subject – contains Math, Reading, and Writing
- Score – contains the corresponding marks

This transformation makes it easier to analyze and compare student performance across different subjects.

11.Creating DAX Measures

- Average Math Score

Average Math Score = AVERAGE('Student Performance'[Math Score])

- Average Reading Score

Average Reading Score = AVERAGE('Student Performance'[Reading score])

- Average Writing Score

Average Writing Score = AVERAGE('Student Performance'[Writing score])

- Overall Average Score

Overall Pass % =

DIVIDE(

CALCULATE(

COUNTROWS('Student Performance'),

'Student Performance'[Pass/Fail] = "Pass"

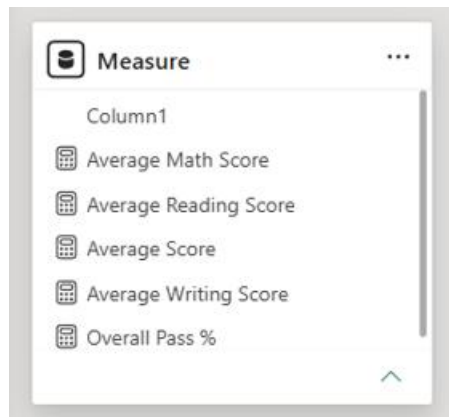
),

COUNTROWS('Student Performance'),

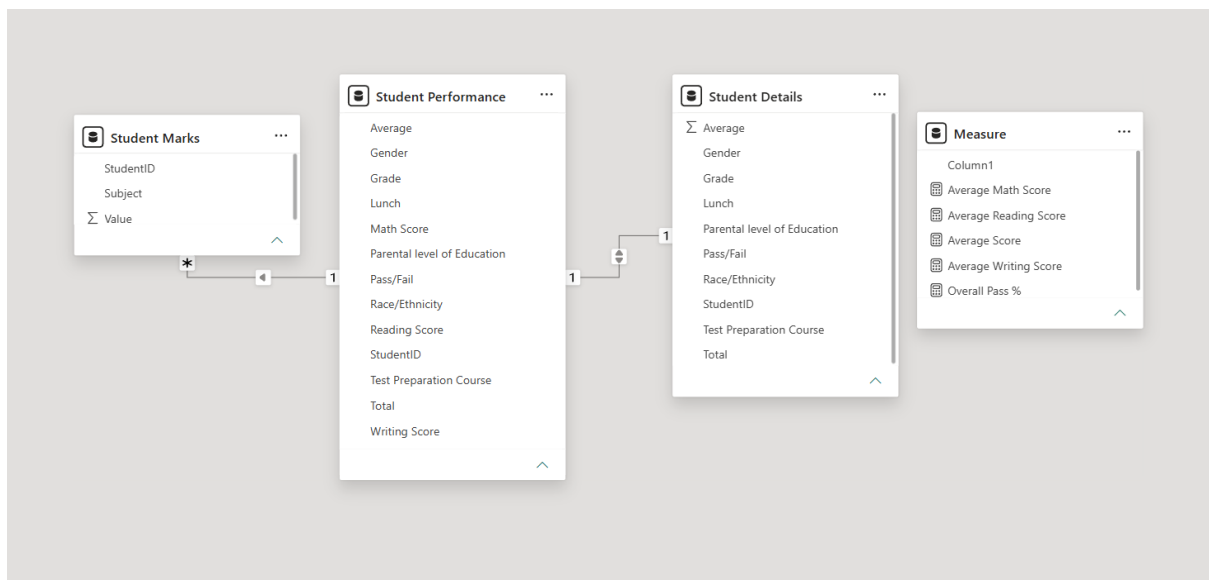
0

)

- Overall Pass %



12.Final Data Model Relationship



13.Conclusion

In this Week 7 project, I used Power BI Power Query to transform the Student Performance dataset. I added StudentID, reordered and renamed columns, created Total and Average scores, created Grade and Pass/Fail classifications, unpivoted the subject scores, and rounded the Average Score.

This project helped me understand how to use Power Query for data cleaning and transformation and how to prepare data for further analysis and visualization in Power BI.